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Linear Algebra : Part-2

Note: Here in this part, we'll learn about Metrices and its different operations.
Moreover, in this part we'll only learn about Metrices and in next part, we'll have the intution of what we learn today.

1) Types of Metrices.

1) Row matrix :

$$\begin{bmatrix} 2 & 4 & 5 & 7 \end{bmatrix}$$

2) Column matrix.

$$\begin{bmatrix} 1 \\ 4 \\ 8 \\ 7 \end{bmatrix}$$

3) Square matrix : No. of rows = No. of cols.

$$\begin{bmatrix} 2 & 4 \\ 3 & 9 \end{bmatrix}_{2\times 2}$$ (2 rows and 2 cols)

In square matrix, elements at ann are called diagonals.

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ 7 & 9 & 8 \end{bmatrix}_{3\times 3}$$

3) Diagonal matrix : ~~All~~ non A square matrix whose all non-diagonal elements are zero.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}_{3\times 3}$$

Topics covered.

1) Types of metrices
2) Matrix equality
3) Scaler operations
4) Matrix addition, subtraction and multiplication
5) Transpose of a matrix
6) Determinant
7) Minor
8) Co-factor
9) Adjoint
10) Inverse of a matrix
11) Solving a system of linear equations.

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4) Scalar matrix : A square matrix whose diagonal element are equal. and other elements are zero.

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}_{3\times3}$$

5) Identity matrix : A square matrix whose diagonal elements are 1 and other are zero.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3}$$

6) Zero matrix : A matrix whose all elements are zero.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2\times3}$$ This is a 2×3 zero matrix.

7) Ones matrix : A matrix whose all the elements are ~~one~~ 1.

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}_{2\times3}$$ This is a 2×3 ones matrix.

2) Matrix equality.

1) If shape of matrices are same and
2) All the elements of both the matrices are same.

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \end{bmatrix}_{2\times3} \qquad B = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \end{bmatrix}_{2\times3}$$

Here $A = B$.

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3) Scalar operations :

There're two types of scalar operations.
i) Addition / subtraction
ii) Multiplication.

$$A = \begin{bmatrix} 2 & 8 \\ 7 & 3 \end{bmatrix} \qquad K = 2$$

- $A + K = 2 + \begin{bmatrix} 2 & 8 \\ 7 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 9 & 5 \end{bmatrix}$

- $A - K = 2 - \begin{bmatrix} 2 & 8 \\ 7 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 6 \\ 5 & 1 \end{bmatrix}$

- $A \cdot K = 2 \begin{bmatrix} 2 & 8 \\ 7 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 16 \\ 14 & 6 \end{bmatrix}$

$\Rightarrow$ Multiplying a matrix with (-1) gives negative of that matrix.

- $A \times (-1) = -1 \begin{bmatrix} 2 & 8 \\ 7 & 3 \end{bmatrix} = \begin{bmatrix} -2 & -8 \\ -7 & -3 \end{bmatrix} = (-A)$

- $-1 \begin{bmatrix} -4 & 6 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 4 & -6 \\ -2 & 3 \end{bmatrix}$

- If A, B and C are matrices,
  $AB + BC = A(B+C)$ ~~$= (A+B)C$~~

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4) Matrix Addition, subtraction and Multiplication.

i) Matrix addition :

- Shape of all the matrices should be same.
- The resultant matrix will be of same shape.

$$\begin{bmatrix} a1 & a2 \\ a3 & a4 \end{bmatrix}_{2\times2} + \begin{bmatrix} b1 & b2 \\ b3 & b4 \end{bmatrix}_{2\times2} = \begin{bmatrix} a1+b1 & a2+b2 \\ a3+b3 & a4+b4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2+1 & 5+4 \\ 7+3 & 9+2 \\ 10 & 11 \end{bmatrix}$$

- $A + B = B + A$
- $A + (B + C) = (A + B) + C$ (Associative rule)

ii) Matrix subtraction :

- There is nothing like matrix substraction.
- We basically combine scalar multi and matrix addi to perform matrix subtraction.
- We perform $A + (-1)B$

$$A = \begin{bmatrix} 2 & 5 \\ 7 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} \Rightarrow -1\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -1 & - \\ -3 & - \end{bmatrix}$$

$$\therefore A - B = \begin{bmatrix} 2-1 & 5-4 \\ 7-3 & 9-2 \\ 4 & 7 \end{bmatrix}$$

- This is how it is mathematically but we can simply do $A - B$

$$\begin{bmatrix} 2 & 5 \\ 7 & 9 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 4 & 7 \end{bmatrix}$$

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A) Additive Identity

- $A + X = A$
- For example

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}_{2\times2} + X = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}_{2\times2}$$

- We know that X should be of same shape then only addition can be performed.
- $X = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times2}$ (Zero Matrix)
- So, for any matrix, an additive identity is always a zero matrix of same size.

B) Additive Inverse.

- $A + X$ = Zero matrix.
- For example

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}_{2\times2} - X_{2\times2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times2}$$

- Obviously X should be of same shape.
- $X = \begin{bmatrix} -2 & -3 \\ -4 & -5 \end{bmatrix}_{2\times2} = -A$
- So for any matrix, an additive inverse is always the negative matrix of that particular matrix.

5) Matrix Multiplication.

Rules:

i) Multiplication of Matrices is only possible if no. of columns of A = no. of rows of B. $n = m$

ii) The resultant matrix is always of size no. of rows of A X no. of columns of B. $m \times n$

For example:

$$\begin{bmatrix} A1 & A2 & A3 \\ A4 & A5 & A6 \\ A7 & A8 & A9 \end{bmatrix} \times \begin{bmatrix} B1 & B2 & B3 \\ B4 & B5 & B6 \\ B7 & B8 & B9 \end{bmatrix} = \begin{bmatrix} A1B1 + A2B4 & A1B2 + A2B \\ A4B1 + & A4B2 \\ A5B & A5B \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times3} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3\times3}$$

Resultant matrix = 2×3

$3 = 3$ possible

Resultant Matrix

$$= \begin{bmatrix} 1+8+21 & 2+10+24 & 3+12+27 \\ 4+20+42 & 8+25+48 & 12+30+54 \end{bmatrix}_{2\times3}$$

$$= \begin{bmatrix} 30 & 36 & 42 \\ 66 & 81 & 96 \end{bmatrix}$$

→ Some important points.

- $AB \neq BA$

For example: $A=(2,3)$ and $B=(3,3)$

$A \cdot B$ is possible but $B \cdot A$? $B=(3,3)$ $A=(2,3)$.

So $B \cdot A$ is not possible. Hence $AB \neq BA$.

- Associative rule is applicable. $(AB)C = A(BC)$
- $AB + AC = A(B+C)$

→ Multiplicative Identity.

- $AI = A$.
- For example

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}_{2\times2} \cdot I = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

- $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times2}$ Identity matrix

- So for any matrix, multiplicative identity is always an identity matrix. Hence, $\boxed{AI = A}$.

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6) Transpose of a Matrix.

Row becomes column and column becomes row

For example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

7) Determinant of a Matrix.

i) $A = [4]_{1\times1}$. $\det(A) = 4$

2) $B = \begin{bmatrix} 2 & 3 \\ 8 & 7 \end{bmatrix}_{2\times2}$. $\det(B) = (2\times7) - (8\times3) = 14 - 24$
$= \underline{(-10)}$

3) $C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. $\det(C) = 1\times\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$

$= 1(45-48) - 2(36-42) + 3(32-35)$
$= 1(-3) - 2(-6) + 3(-3) = -3 + 12 - 9$
$\therefore \det(C) = 0$ (singular matrix)

Note: A matrix whose determinant is zero is called a singular matrix

8) Minor.

- Every element of a matrix will have its own minor.
- To find the minor of an element, delete its row and column and find the determinant of the other elements. This is the minor of that element.
- It is denoted by M_{ij}.

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1) $A = \begin{bmatrix} 3 & 4 \\ 8 & 7 \end{bmatrix}_{2\times 2}$

- Minor (3) = 7 , Minor (4) = 8
- Minor (8) = 4 and Minor (7) = 3

2) $B = \begin{bmatrix} 1 & 5 & 6 \\ 3 & 7 & 2 \\ 4 & 8 & 9 \end{bmatrix}$

- Minor (1) = $7\times9 - 8\times2$ = 47 , Minor (5) = $(3\times9)-(4\times2)$ = 19
- Minor (6) = (-4), Minor (3) = (-3), Minor (2) = (-12)
- Minor (4) = (-32), Minor (8) = (-16), Minor (9) = (-8)

- We define minor for a matrix and not a scalar
- Even if we try to find the minor of a scalar we delete row and column, so that single elements gets deleted and remains an empty matrix
- Determinant of an empty matrix is 1 so minor of a matrix is 1.

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9) Cofactor.

- Cofactor is denoted by A.

Formula:

$A_{ij} = (-1)^{i+j} * M_{ij}$

$B = \begin{bmatrix} 1 & 5 & 6 \\ 3 & 7 & 2 \\ 4 & 8 & 9 \end{bmatrix}$

1 =

- $A_{11} = (-1)^{1+1} + M_{11} \Rightarrow 47$ $(7\times9)-(8\times2)$ $\quad 1\times47 = 47$
- $5 = A_{12} = (-1)^{1+2} + M_{12} = 19$ $(3\times9)-(4\times2)$ $\quad = -1 * 47 = (-47)$
- $6 = A_{13} = (-1)^{1+3} + M_{13} = (-4)$ $(8\times3)-(4\times7)$ $\quad = (-1) * (19) = (-19)$
- $\quad = 1 * (-4) = (-4)$

Similarly, find co-factor of other elements.

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10) Adjoint of a matrix.

$$B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad adj(B) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T;$$

where A = Cofactor.

$$B = \begin{bmatrix} 1 & 5 & 6 \\ 3 & 7 & 2 \\ 4 & 8 & 9 \end{bmatrix} \Rightarrow adj(B) = \begin{bmatrix} 47 & -19 & -4 \\ 3 & -15 & 12 \\ -32 & 16 & -8 \end{bmatrix}^T = \begin{bmatrix} 47 & 3 & -32 \\ -19 & -15 & 16 \\ -4 & 12 & -8 \end{bmatrix}$$

$A_{11} = (-1)^2 \times 47 = 47$
$A_{12} = (-1)^3 \times 19 = (-19)$
$A_{13} = -1^4 \times -4 = (-4)$
$A_{21} = -1^3 \times -3 = 3$
$A_{22} = -1^4 \times -15 = (-15)$
$A_{23} = -1^5 \times -12 = 12$
$A_{31} = -1^4 \times -32 = (-32)$
$A_{32} = -1^5 \times -16 = 16$
$A_{33} = -1^6 \times -8 = (-8)$

11) Inverse of a matrix.

- An inverse matrix is a matrix which when multiplied with the original matrix, results into an identity matrix.
- $A \cdot A^{-1} = I$.

- The inverse matrix is defined only for square matrices.
- Not all square matrices have an inverse.
- A matrix is invertible only if it is non-singular, meaning its determinant is non-zero.
- If the determinant of A is zero, it is a singular matrix and it doesn't have an inverse.

- Applications of inverse matrix are solving system of linear equations, find the solution of matrix eqn and performing various matrix operations.

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Formula:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

- $(AB)^{-1} = A^{-1} \times B^{-1}$

12) Solving a system of linear equation.

$x + y = 5$ and $4x + 3y = 15$

$x = y - 5$

$4(y-5) + 3y = 15 \Rightarrow 4y - 20 + 3y = 15$

$\boxed{x = 5 - 5 = 0}$

$7y = 35 \Rightarrow \boxed{y = 5}$

Now lets solve this using matrix's method.

$$\begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \end{bmatrix} \Rightarrow AX = B.$$

$$\boxed{X = A^{-1} B}$$

A X B

$$A^{-1} = \frac{1}{\det(A)} (\text{adj } A)$$

- $\det(A) = 3 - 4 = (-1) \rightarrow$ Non-singular matrix. So inverse possible

- $\text{adj}(A) = [\text{cofactor}]^T$

$$= \begin{bmatrix} 3 & -4 \\ -1 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$$

$-1^{1+1} \times 3 = 3$

$-1^{3} \times 4 = (-4)$

$-1^{2+1} \times 1 = (-1)$

$-1^{2+2} \times 1 = 1$

$$\therefore A^{-1} = \frac{1}{-1} \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix}$$

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$$\Rightarrow X = A^{-1} B$$

$$= \underset{2\times2}{\begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix}} \underset{2\times1}{\begin{bmatrix} 5 \\ 15 \end{bmatrix}} = \begin{bmatrix} -15+15 \\ 20+(-15) \end{bmatrix} \therefore \boxed{X = \begin{bmatrix} 0 \\ 5 \end{bmatrix}}$$